

PACT @ Ecosperity Week 2026 – Q&A

Written responses to audience questions

Partnership for Carbon Transparency (PACT) at WBCSD | May 2026

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Event 1: Carbon Data Is Becoming Economic Currency: Is Asia Ready?

Q1: How does PACT handle confidentiality when suppliers don't want to reveal lower-level / process-level data?

PACT is based on a peer-to-peer, consent-based exchange: a supplier shares a PCF only with the specific buyer they choose, not publicly. PACT does not own or store any of the data. Suppliers are not required to disclose raw process data or facility-level figures, only the aggregated cradle-to-gate PCF value alongside methodology metadata. The PACT Technical Specifications (for PCF Exchange) include data quality indicators and primary data share metrics, which give receivers confidence in data reliability without exposing underlying processes.

More on the PACT data model and confidentiality approach: [PACT Methodology \(V3.0\)](#)

Q2: How does PACT deal with data quality gaps when suppliers do not yet have mature carbon-accounting capabilities?

The PACT Methodology V3.0 addresses supplier data quality gaps through a structured, transition-oriented approach rather than strict exclusion or substitution. PACT recognizes that many suppliers lack mature carbon accounting systems and therefore allows the use of secondary data and proxy emission factors within a defined data hierarchy, while progressively steering companies toward higher-quality primary data. Data quality is managed through a formal Data Quality Indicator (DQI) framework and Primary Data Share (PDS), which together quantify both the representativeness and completeness of underlying data and create transparency about data maturity levels across the value chain.

Where supplier-specific data is not yet available, PACT enables fallback use of secondary datasets and industry averages, but requires clear disclosure and structured scoring of data quality so that

gaps remain visible and comparable rather than hidden. Over time, the methodology drives improvement by requiring reporting of both PDS and DQR, encouraging suppliers and buyers to systematically upgrade data quality rather than rely indefinitely on estimates.

PACT's Usability Model is also deliberately tiered, it is fit-for-purpose, not all-or-nothing. Suppliers can start with spend-based or industry-average data (Level 1) and progress to primary, supplier-specific data (Level 3) over time. Each PCF exchanged through PACT carries explicit data quality indicators (Data Quality Rating, Primary Data Share) so the buyer knows exactly what they are receiving. This transparency around data quality is itself the starting point for improvement.

See the PACT Usability Model: [Using PCF Data in Practice](#)

Q3: Where do EPDs fit in with product footprint requirements and digital product passports? Are they becoming less necessary?

EPDs are not becoming less necessary, they serve a distinct purpose. An EPD is a full, multi-impact LCA document (covering GWP, water, toxicity, etc.) governed by ISO 14025, primarily used for public disclosure, green building certification (LEED, BREEAM), and procurement. A PCF is a narrower instrument: GHG emissions only, machine-readable, designed for supply chain data exchange.

The relationship is complementary: the GWP module inside an EPD contains the carbon data that a PCF would express. PACT is actively exploring how EPD data can serve as a verified source for PACT-compatible PCFs, effectively making EPD investments travel further into digital supply chains.

On Digital Product Passports (DPPs): PACT's open API infrastructure is directly relevant here. PCF data exchanged via PACT-conformant solutions can become a core data input into DPPs, particularly under the EU's Ecodesign for Sustainable Products Regulation (ESPR).

Q4: Singapore has pushed back mandatory Scope 3 reporting. Does this mean there is less urgency for carbon data exchange?

From PACT's perspective, the regulatory signal is one driver, but the commercial pull is growing independently. Global institutional buyers, major developers, REITs, infrastructure funds, and multinationals, are already requesting PCF data from suppliers as part of their own Scope 3 calculations and net-zero commitments. Singapore-based suppliers serving these buyers face data requests regardless of local mandatory timelines.

Additionally, the EU's CBAM, CSRD, and ESPR create extraterritorial demand: any company exporting to or sourcing from Europe will need product-level carbon data. The Singapore delay changes the domestic compliance timeline, not the trade and finance-driven urgency.

For resources available to business, you can access Singapore's free digital platform designed to empower businesses in their decarbonisation journey: [Home | NetZeroHub.SG](#)

Q5: What are the implementation challenges for industries shifting to PACT methodology? How does PACT coexist with other measures?

Key implementation challenges include:

1. Data collection maturity, many suppliers lack granular primary data
2. Organisational capacity, PCF calculation requires cross-functional effort across procurement, operations, and sustainability
3. Systems integration, connecting PCF data flows to ERP and procurement systems
4. Awareness and trust, both buyers and suppliers need to understand what they are requesting and providing.

On coexistence: PACT is designed as infrastructure, not a competing standard. It is designed to complement and operationalize existing international standards rather than replace them. The methodology is built on established LCA and carbon accounting frameworks such as ISO 14067, while providing additional guidance needed for interoperable, exchange-ready product carbon footprint (PCF) data across value chains.

PACT also aligns with sector-specific initiatives (e.g. TfS for chemicals, GLEC for logistics, etc., see full list [here](#)), and emerging regulatory frameworks by offering a practical, harmonized approach for PCF calculation, data quality assessment, and digital data exchange. This enables companies to leverage existing investments in standards compliance while improving consistency, comparability, and scalability of emissions data sharing across industries.

See how PACT aligns with existing standards: [PACT Methodology overview](#)

Q6: How can we secure the trustworthiness of carbon data without incurring too much cost for monitoring and verification?

PACT's V3 Methodology (Section 5) is built precisely around the cost-versus-trust trade-off. Rather than requiring verification of every individual PCF (accurate but costly at scale) it requires verification at the **calculation-model** level in the short term, and at the **PCF Program** level from 2030. The Methodology frames this as "*a simplified verification approach that balances rigor with practicality*": verifying the system that generates PCFs, once, rather than each output, is what keeps assurance affordable as volumes grow.

Carrying that trust through the exchange is the job of the **PACT Technical Specifications**, which require data quality metadata — Primary Data Share, Data Quality Rating, and assurance level — to travel with every PCF. This lets a receiver gauge confidence in a number without needing full third-party verification of it, so trust scales with the data rather than with audit cost.

Innovation is where the cost curve really bends: AI-assisted data extraction, automated verification, and interoperable audit trails are areas where new entrants can meaningfully lower the cost of trust.

See PACT Methodology V3, Section 5: [Download here](#)

Q7: Is the push for carbon data more regulatory-heavy, or is there a genuine business case without external intervention?

Both are present, and they are reinforcing. On the regulatory side: CBAM, CSRD, ESPR, and green procurement requirements in multiple jurisdictions are creating hard compliance timelines.

On the business side: companies with better PCF data are able to make smarter procurement decisions, demonstrate climate performance to investors and customers, and respond faster to due diligence requests.

The clearest near-term business case is risk management: companies without granular Scope 3 data are increasingly disadvantaged in tender processes, financing discussions, and customer relationships. PCF data is becoming a commercial requirement in certain sectors (chemicals, automotive, electronics) regardless of regulation.

Q8: How does PACT differentiate from Carbon Measures (CM) and ISO-GHGP convergence?

PACT is infrastructure. Its job is to enable trusted product-level sustainability data to flow reliably across supply chains, systems, and borders. That means it can serve many different use cases, including carbon transaction accounting approaches like Carbon Measures.

Carbon Measures is still an emerging initiative, and frankly it's not yet fully clear what the final deliverable will look like. We're in early conversations with the team behind it, and we'll make sure PACT stays plugged into that work. Our commitment, as always, is to support the market in accelerating transparency, wherever the methodological innovation leads.

On ISO and GHG Protocol convergence: PACT is actively involved in the ISO14067 update process, and our community is part of that conversation. Our role is to operationalize whatever sustainability methodology the market demands, not to drive methodological convergence ourselves. When the market aligns on standards, PACT is the infrastructure that makes sure the data can actually flow.

PACT Methodology V3 (released April 2025): [Download here](#)

Q9: Given that PCFs are data-heavy, costly, and short-lived, where can new players innovate to accelerate PCF generation and lower costs?

Several high-value areas for innovation exist for solution providers:

1. **Automated data extraction** from existing systems (ERP, MES, utility bills) to reduce manual effort at the point of data collection
2. **LCA database integration** that pre-populates secondary emissions factors while flagging where primary data would most improve accuracy
3. **AI-assisted activity and process data capture**, going beyond secondary data gap-filling to intelligently extracting and structuring the underlying operational data that drives PCF calculations

4. **Ready application of sector-specific rules and normative data**, reducing the expertise barrier by embedding PCR requirements, allocation rules, and sector benchmarks directly into the calculation workflow
5. **Shared model structures across multiple products**, enabling companies to build once and scale across a product portfolio rather than configuring each PCF from scratch
6. **Automated assignment of Data Quality Indicators (DQIs)** throughout the calculation model, removing a step that currently requires significant manual judgement
7. **Automated proposal of reduction initiatives** based on process and geographic context, marginal abatement costs, and available technologies — turning PCF generation from a reporting exercise into an actionable decarbonisation tool
8. **Verification-as-a-service** models that spread the cost of third-party assurance across supplier networks

The PACT-conformant solution ecosystem already includes 50+ technology providers building in this space. New entrants can differentiate on sector depth, ease of onboarding, and cost per PCF, but the real opportunity is in moving PCFs from static data points to live decision-support infrastructure.

Explore PACT-conformant solutions, [here](#), or access our simple guide on how to join it, [here](#).

Q10: Where is the biggest friction point preventing broader adoption of an interoperable carbon data ecosystem?

The biggest friction is the asymmetry of effort and incentive: the supplier bears the cost of calculating and sharing PCF data, while the buyer captures most of the value. Until buyers create clearer commercial signals (procurement preferences, pricing differentials, contractual requirements), many suppliers will deprioritise PCF generation due to resource and capability constraints.

The second major friction is fragmentation: different buyer requests, different methodologies, different formats. This is precisely what PACT's interoperability layer is designed to solve: a supplier calculates once and can share with multiple buyers in a standardised format, rather than responding to bespoke data requests repeatedly.

Q11: What is the ROI for PCF disclosure and exchange?

The ROI case has several components:

1. Reduced Scope 3 data collection costs over time: once a PCF infrastructure is in place, marginal cost of sharing falls sharply
2. Access to green procurement contracts: several major buyers (in automotive, consumer goods, chemicals) are making PCF data a condition of supply
3. Reduced compliance cost: a single PACT-based PCF can satisfy multiple regulatory and customer disclosure requirements simultaneously
4. Decarbonisation insight: granular PCF data reveals where emissions reductions have the highest impact, enabling smarter capital allocation.

You can access PACT's latest White Paper which links PCF data to business use cases (the PACT Usability Model): [Using PCF Data in Practice](#)

Q12: The burden of demonstrating trust for carbon data exchanged lies with which party?

PACT distributes trust responsibilities across the ecosystem rather than placing it entirely on one party. The supplier is responsible for calculating the PCF correctly and providing accurate methodology metadata (per PACT V3 requirements). The solution provider (software platform) is responsible for implementing the PACT Technical Specifications correctly and achieving PACT conformance (after passing the Conformance Process, they officially become part of the [PACT Network](#)). The PACT network provides the identity management and directory service that establishes trust connections between solutions. Third-party assurance bodies provide external verification of PCF calculation models.

The buyer retains responsibility for assessing the data quality indicators provided and determining fitness for their specific use case, which is why [PACT's Usability Model](#) is so important: it gives buyers a framework for making that assessment.

Q13: What is the Singapore government's direction on green procurement beyond GreenGov.sg?

This is best addressed by the Singapore government directly. For the latest public guidance on green procurement policy, we recommend checking GreenGov.sg: [Home | Singapore Green Plan 2030](#)

PACT is engaging with the Singapore Business Federation (SBF) on how product-level carbon data infrastructure can support Singapore's ambitions. We will share more as this engagement develops.

For more resources, access Singapore's free digital platform designed to empower businesses in their decarbonisation journey: [Home | NetZeroHub.SG](#)

Q14: Is PACT more geared towards manufacturing? How about construction, which has a large and rising footprint?

PACT is explicitly industry-agnostic by design: the methodology and technical specifications apply to any product category. While early adoption has been strongest in manufacturing-intensive sectors (chemicals, electronics, automotive), the built environment is another focus area for PACT expansion.

The construction sector faces a specific challenge: it has a mature EPD ecosystem but lacks the bilateral exchange infrastructure to flow that data into Scope 3 calculations. Bridging EPDs and PCFs for the built environment would essentially make EPD investments interoperable with PACT's exchange layer. This would open the PACT infrastructure to steel, cement, glass, insulation, and other construction materials.

We see Singapore, with its strong Green Mark and BREEAM adoption, as a natural starting point for a regional built environment working group. To date, PACT does not have specific working group on this topic. However, if you would like to join PACT and become a sponsor, please contact us: pact@wbcsd.org

Q15: Is PACT considering opening up the data platform for wider transparency use cases and policy impact?

PACT is not a platform, it is a peer-to-peer network of interoperable solutions. The PACT Technical Specifications are an API specification which is open for any software solution to implement, and the methodology is free to use.

PACT is not a closed platform, it is an open and neutral infrastructure, any solution provider or organisation can build on it.

For broader transparency and policy use cases (e.g., public PCF databases, carbon benchmarks, regulatory reporting feeds), the question of who operates such infrastructure and under what governance is being discussed within the PACT community.

PACT's role is to ensure that any such infrastructure uses a common, interoperable data standard for accurate calculation and seamless exchange of this data across systems and borders.

Q16: What is the Singapore government's view on operational vs embodied carbon?

This is a question best directed to BCA (Building and Construction Authority) and NEA (National Environment Agency). Singapore's Green Building Masterplan and the Singapore Green Building Council's work on whole-life carbon are the primary public references.

PACT's current focus is on embodied carbon (product-level, supply-chain emissions), the area where PCF data is most directly relevant.

Q17: What is the roadmap to move from ~5,000 suppliers to significantly broader adoption?

PACT's scale-up strategy works on both sides of the market simultaneously.

On the **demand side**: regulation (e.g. CBAM, ESR, green procurement mandates like Singapore's GreenGov), buyer demand for PCF data in procurement, R&D, and Scope 3 hotspotting, and finance mechanisms (e.g. green loans, ESG ratings, CSRD) are all creating pull that didn't exist five years ago.

On the **supply side**: PACT recognizes the role of software providers to enable standardised PCF exchange at scale, service providers (consultancies, auditors) to build implementation capacity, training programs to upskill practitioners, and regional enablement partners to localise and implement PACT in specific markets, China, Singapore, and Japan among them.

Underpinning both is PACT's role as **neutral, open infrastructure**. Rather than treating fragmentation as a problem to solve, PACT turns it into an asset, different systems, standards, and

organisations can plug into PACT's common methodology and technical standard, making their data interoperable without changing their own approaches. The result is a common layer that connects a fragmented landscape rather than replacing it. The trajectory is not linear, it accelerates as more players on both sides of the market join the same infrastructure.

Q18: As carbon pricing strengthens, how does Singapore ensure industries are incentivised to decarbonise without carbon leakage?

Carbon leakage prevention through pricing design is primarily a government policy question — Singapore's carbon tax trajectory and sector coverage are the key instruments, and we recommend engaging MAS, NEA, and EDB directly on this.

From PACT's perspective, granular product-level carbon data is a prerequisite for any effective carbon border adjustment mechanism: you cannot assess leakage risk or apply border adjustments fairly without reliable PCF data at the product level. PACT's infrastructure is exactly the kind of data layer that makes such policy mechanisms administrable.

Q19: How is PACT aligning with EcoVadis or CDP?

PACT is complementary to both. EcoVadis assesses overall ESG performance at the company level. CDP captures corporate-level climate disclosures. PACT operates at the product level, enabling the granular, supply-chain-specific data that feeds into the corporate disclosures that EcoVadis and CDP collect.

More on PACT's ecosystem alignment: [PACT Methodology page](#)

Q20: How does PACT differentiate from Carbon Measures?

See Q8 above.

Q21: Does Singapore plan to have wider mandatory PCF disclosure on consumer products?

No mandatory consumer product PCF disclosure policy has been announced in Singapore to our knowledge as of mid-2025. This is a question best directed to NEA and MTI. Globally, the EU's ESPR and Green Claims Directive are the leading policy frameworks heading toward mandatory product-level carbon disclosure. Singapore-based companies exporting to Europe should be tracking these closely.

Q22: What is the hardest part operationally when trying to obtain trusted emissions data across large supply chains (e.g., Swire Coca-Cola)?

The hardest operational challenges that are flagged by PACT members are consistent across large supply chains:

1. **Supplier onboarding:** getting thousands of suppliers to calculate and share PCF data in a consistent format requires significant relationship management, upskilling and technical support;
2. **Data quality heterogeneity:** suppliers at very different levels of carbon accounting maturity produce data of widely varying reliability
3. **Systems integration:** connecting PCF data flows into procurement and Scope 3 accounting systems
4. **Change management:** internal alignment across procurement, sustainability, and finance teams.

PACT's interoperability standard directly addresses point 3. Points 1-2-4 are the operational transformation challenges that consultants and solution providers in the PACT ecosystem are helping companies work through.

Q23: How does PACT ensure data robustness and comparability given variability across industries and sectors?

PACT addresses this through three mechanisms:

1. **A common methodology** (PACT V3) that specifies calculation rules, boundary definitions, allocation methods, and data quality requirements applicable across all sectors
2. **Mandatory data quality metadata exchanged alongside every PCF** — Primary Data Share, Data Quality Rating, assurance level, and reference period — so receivers can assess what they are getting
3. **Sector-specific data model extensions** that provide additional guidance within the common framework for specific product categories (chemicals, aluminium, logistics, etc.), ensuring sector-relevant detail without sacrificing cross-sector comparability.

Comparability is a key topic for our community, and we will focus on this within PACT's Use Case Forum with our members. If you want to get involved, please contact pact@wbcsd.org.

Q24: Climate change involves more than GWP/carbon. How does PACT include other impact indicators?

PACT V3 focuses on greenhouse gas emissions (CO₂e) as its primary scope. This is intentional, to establish a clear, comparable, interoperable standard for the most critical climate metric. Other environmental indicators (water, biodiversity, toxicity, resource depletion) are covered by full LCA and EPDs, not by PCFs as currently defined. This aligns with international standards such as ISO 14067, which also defines CFPs around the single impact category of climate change.

PACT acknowledges this is a limitation and the broader question of multi-impact product-level data exchange is an area of ongoing discussion within the PACT ecosystem. For built environment specifically, EPDs already cover this broader picture — which is another reason why bridging EPDs and PCFs is strategically important.

Q25: How does the PACT methodology capture emissions from trading?

PACT calculates cradle-to-gate PCFs, emissions from raw material extraction through to the factory gate of the declaring company. This includes all upstream supply chain emissions attributable to the product. Trading activity (buying and reselling goods without transformation) is not a standard PCF scenario in PACT's current scope; the methodology is designed for manufacturers and producers calculating the footprint of their own products. For financial institutions and traders tracking embodied emissions in traded commodities, PACT's product-level data provides the underlying carbon intensity figures that commodity-level accounting frameworks can use.

Q26: How does PACT address methodological differences across sectors and industries?

Through a two-layer architecture: a common core methodology (PACT V3) that sets universal rules for boundary definitions, allocation, data quality, and exchange; and sector-specific extensions (Partnership for Carbon Accounting Frameworks, or PCAF-style sector guidance) that add product-category-specific calculation rules on top of the core. This means a steel PCF and a chemical PCF are comparable at the level of their GWP values, even if the detailed calculation rules differ at the process level. PACT works with sector bodies to develop and validate these extensions.

Q27: Are there regulatory initiatives to encourage PCF adoption?

Yes, and the landscape is moving fast. Key regulatory drivers: EU CBAM (Carbon Border Adjustment Mechanism), requires carbon intensity data for imports in steel, cement, aluminium, fertilisers, and electricity. EU ESPR / Digital Product Passport — mandates product-level environmental data for a growing list of product categories. EU CSRD — requires Scope 3 disclosure, for which PCF data is the most granular input. Singapore Carbon Tax — as pricing increases and potentially expands to cover more sectors, product-level carbon data becomes increasingly material.

PACT's PCF standard is designed to be compatible with these regulatory frameworks, meaning a PACT-conformant PCF can satisfy multiple disclosure requirements simultaneously.

Event 2: Building the Carbon Data Network: From Interoperability to Implementation

Q1: EPDs and PACT both use LCA methodologies. EPD GWP numbers are verified. How does PACT get its LCA verified?

EPDs are verified by accredited third parties at the product-result level, rigorous, and a real strength. PACT V3 verifies at a different level: not each individual PCF, but the **PCF Calculation Model** (short term, limited assurance) and, from 2030, the **PCF Program** that governs how PCFs are generated. The Methodology calls this "*a simplified verification approach that balances rigor with practicality*", a deliberate choice to phase verification in and meet companies where they are today.

The two are complementary. PACT's "existing verification" provision (Section 5.6.1) lets verification done for other purposes count toward PACT's requirements, so an EPD's assurance can feed into a PACT exchange — and a company starting with PACT builds the data discipline that makes fuller product-level verification more attainable over time. Because the two assure at different levels, the mapping isn't automatic: a PACT verification working group is currently exploring how to bridge them, with operational guidance forthcoming.

To get involved, contact pact@wbcsd.org. See PACT Methodology V3, Section 5, [here](#).

Q2: How do we move from just steel and concrete to cover the entire scope of construction?

Steel and concrete are the right starting point — they typically represent 60-80% of embodied carbon in structural construction. But full scope coverage requires extending to: MEP systems, facades and glazing, insulation and fitout materials, and infrastructure components.

The pathway suggested during the presentations is:

1. Establish PCF data flows for the high-impact materials first, building supplier data capability and buyer demand signals
2. Use EPDs as the data source for categories with mature EPD programs, bridged into PACT exchange format
3. Build sector-specific PACT extensions for construction product categories without existing EPD programs. This is a multi-year effort that requires collaboration between PACT, EPD program operators (International EPD System, IBU, ECO Platform), and construction sector bodies.

Q3: Can you share a case study on decarbonisation or PCF exchange that delivers business ROI without relying on regulation?

Several PACT member companies have documented business-driven cases. In the chemicals sector, BASF and Dow have used PCF data to differentiate products commercially — buyers in the automotive and consumer goods sector are willing to pay a premium for lower-carbon inputs

when they have verified data to back the claim. In the electronics sector, PCF data has been used to optimise sourcing decisions and reduce supply chain carbon intensity ahead of regulatory requirements. For the built environment specifically, the clearest commercial case is green building certification: projects targeting LEED Platinum or BREEAM Outstanding increasingly need verified embodied carbon data that PCFs provide, and contractors who can demonstrate low-embodied-carbon supply chains win more of these projects.

PACT member case studies: [Case Studies](#)

Q4: Since ~80% of products don't have EPDs, how do you calculate project LCA and make decisions?

In the absence of EPDs or primary supplier PCF data, project LCA relies on generic database values (ecoinvent, GaBi, ICE database, EC3's benchmark data). These are less accurate but usable for early-stage design decisions and carbon benchmarking. The practical approach is tiered: use generic data for low-impact or hard-to-source materials, prioritise primary data collection for the top 3-5 materials by carbon contribution.

PACT and EPD programs are part of the solution to the 80% gap — as more suppliers publish EPDs and adopt PCF reporting, the coverage improves. In the interim, tools like the EC3 database (Building Transparency) and OneClickLCA aggregate available EPD data and allow substitution of generic data where EPDs exist.

Q5: Why does a construction company need to buy physical green steel rather than just green steel certificates?

This is a live debate in the sector. Green steel certificates (similar to RECs in energy) allow buyers to claim low-carbon steel without physically receiving it. The key arguments against certificate-only approaches:

1. **Additionality concerns:** certificates may not drive new green steel production
2. **Scope 3 accounting rules:** under the GHG Protocol market-based method, certificates can be used for Scope 3, but many buyers and standard-setters (including SBTi in some guidance) prefer physical traceability
3. **Customer and certification requirements:** LEED and BREEAM projects often require documented supply chain provenance, not just certificate claims.

This is partly a regulatory gap rather than a pure policy blockage. As green steel standards and certificate registries mature, the certificate route may become more accepted. For now, physical procurement may be a more defensible approach for high-profile green building projects.

Q6: Does ESGpedia cover end of life and circularity as well?

For detailed information on ESGpedia's platform scope including end-of-life and circularity coverage, please contact ESGpedia directly. Their team can provide the most current and accurate information on platform capabilities. Please contact ESGpedia: Benjamin Tan (Benji) benjamin.tan@esgpedia.io.

Q7: What were the key challenges in doing the EPD project with Integrated Precast?

This is best addressed by the team involved in that specific project. If you'd like an introduction, please reach out to us and we can facilitate the connection.

Q8: Getting companies fully onboard to collate data and understand reporting is a challenge. Can we find better ways to assist?

This is one of the central challenges in scaling PCF adoption. Approaches that are working: (1) Supplier enablement programs run by large buyers, providing calculation tools, training, and data templates; (2) Sector-specific onboarding guides that translate PACT methodology into practical steps for specific product types; (3) Light-touch digital tools that allow suppliers to self-report basic PCF data with guided workflows; (4) Peer learning networks where early adopters in a sector share experience with newcomers. For guidance on how to set up a supplier engagement program, here are some useful resources:

- [Getting started on your journey – how to best engage your suppliers | PACT Insights](#)
- [Design a supplier engagement program - Action Library \(EN\)](#) (WBCSD Climate Drive)
- [Create financial incentives for suppliers - Action Library \(EN\)](#) (WBCSD Climate Drive)
- [Supplier-Engagement-Guidance.pdf](#) (SBTi)

Q9: Product-level targets don't encourage system optimisation (make steel better OR use less). Do you recommend targets like kgCO₂/m²?

Product-level targets (kgCO₂/tonne of steel) and system-level targets (kgCO₂/m² of floor area) serve different purposes and are complementary. Product-level targets drive product decarbonisation — they incentivise manufacturers to reduce production-side emissions. System-level targets (kgCO₂/m²) drive architectural and engineering optimisation — using less material, choosing lower-carbon alternatives, and optimising structural efficiency. We encourage the use of both types of targets to ensure progress across both dimensions.

Additionally, PACT Methodology could be used in principle for cradle to gate type carbon quantification of a building with the declared unit of constructed m². This would mean both dimensions are included in a single metric making any tradeoffs or synergies across product intensity versus constructed building intensity highly visible. Any comparisons across constructed buildings with this metric would need to be approached cautiously to ensure a fair comparison basis.

Q10: How do you engage suppliers to provide carbon data?

Engagement works best when tied to commercial relationships. The most effective approaches:

1. Make PCF data a supplier requirement in RFPs and contracts, buyers with purchasing power create the strongest pull (learn how companies part of the [Sustainable Procurement Pledge](#) approach this)

2. Frame it as a service to suppliers, not just a burden, suppliers with PCF data can differentiate themselves to multiple buyers you need to build the right incentives
3. Provide tools and support to lower the cost of compliance
4. Start with the top 20 suppliers by spend/carbon impact rather than the full supply base.

See resources in Q8 related to supplier engagement.

Q11: How does ESGpedia's decarbonisation simulation engine work? Is it LLM/AI-based or deterministic?

For technical details on ESGpedia's simulation engine architecture, please contact ESGpedia directly. They are best placed to describe their methodology and the role of AI vs. deterministic modelling in their platform (Benjamin Tan benjamin.tan@esgpedia.io).

Q12: Almost no one 'speaks carbon' yet. Do you share details on whether a specific number is good/bad/impossible?

This is a real and important gap. Carbon literacy, understanding what a specific PCF figure means relative to industry benchmarks, is as important as the data itself. PACT's methodology provides the calculation and exchange rulebook needed, but contextualisation requires sector benchmarks.

For construction, resources like the EC3 database (Building Transparency) and the RICS Whole Life Carbon Assessment standard provide benchmark ranges for common materials. For example, global average carbon intensity for structural steel is ~2.0 tCO₂e/tonne; low-carbon electric arc furnace steel can be below 0.5 tCO₂e/tonne. Making these benchmarks visible alongside PCF data is something PACT-conformant platforms are increasingly building in.

For a full view of the sector-specific initiatives that are compatible with PACT, you can visit our website: [PACT Methodology](#)

Q13: Who bears the premium price of the green steel purchase?

Currently, the buyer bears the premium, construction companies or developers who commit to low-carbon specifications absorb the cost differential. The premium varies: low-carbon steel from electric arc furnaces can be cost-competitive in markets with cheap renewable electricity; green hydrogen-based steel still carries a significant premium (estimated 20-50% in most markets today). Over time, as demand scales and policy (carbon pricing, green procurement mandates) internalises carbon costs, the premium should compress. For now, the business case often relies on green building certification premiums, tenant demand, or investor ESG mandates absorbing the cost.

Q14: How and to whom do you charge your service costs? (PACT)

PACT itself (the methodology and technical specifications) is free and open — any company or solution provider can use the PACT methodology and build PACT-conformant solutions at no cost. PACT is a WBCSD-hosted initiative funded by its member organisations and grants.

Costs arise at the solution layer: carbon accounting software, EPD generation services, and PCF calculation platforms charge for their services. For platform-specific pricing, see Q16 below, or visit the [PACT Network](#) page.

Q15: Some construction companies and data centers are sensitive about cost. How to motivate them to decarbonise and use PACT?

Start with the business case that is most relevant to them.

For data centers: regulatory compliance (EU CSRD, Singapore mandatory climate reporting), hyperscaler customer requirements (Microsoft, Google, Amazon all have Scope 3 supplier programs), and green financing access.

For construction companies: green building certification access, institutional buyer requirements, and access to green bond financing for projects with verified embodied carbon performance.

PACT's open calculation and exchange methodology means companies do not need to invest in a single expensive platform, they can start with [PACT-conformant solutions](#) and scale investment as the business case matures.

Q16: How much does it cost to use this platform? (ESGpedia / Smart Tradzt platforms)

For pricing information on ESGpedia's platform, please contact the ESGpedia team directly (Benjamin Tan benjamin.tan@esgpedia.io). For pricing on Smart Tradzt's platform and associated services, please reach out to CK Chung directly for the most accurate and up-to-date commercial information (CK Chung ck.chung@smarttradzt.com).

Q17: Are there any alignments with ESPR and preparing for Digital Product Passports on ESGpedia's platform?

For details on ESGpedia's alignment with the EU Ecodesign for Sustainable Products Regulation (ESPR) and Digital Product Passport (DPP) readiness, please contact ESGpedia directly Benjamin Tan benjamin.tan@esgpedia.io). From a PACT perspective, PCF data exchanged via PACT-conformant solutions is directly relevant to DPP requirements — the PACT API architecture is designed to feed product-level carbon data into broader product data ecosystems, including DPPs.

Q18: Are there examples of PACT-conforming carbon calculations outside of the building sector?

Yes, the built environment is actually a relatively new focus area for PACT. The majority of PACT adoption to date has been in: chemicals (Tfs/BASF/Dow/Covestro), automotive supply chains (electronics, steel, aluminium), consumer goods (Unilever, AB InBev), and logistics (GLEC framework integration). PACT V3 is a sector agnostic methodology designed to be applicable to any product category. See the growing solutions conformant with PACT: [PACT Network](#)

For more information on PACT: carbon-transparency.org | [LinkedIn](#) | Contact: pact@wbcsd.org

For more information on SBF, SEFR & related services: [Home](#) | [NetZeroHub.SG](#) | [Contact Us For SEFR](#)
[| SEFR](#) | [Green 100](#)